



Database Systems

(CS2005)

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Course Instructors

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3057.

Roll No

Section

Final Exam

Total Time (Hrs): 3

Total Marks: 120

Total Questions: 10

Shoaib
Student Signature

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Attempt all the questions.

CLO #1: Explain Fundamental Database Concepts.

1. Answer the following:

- Suppose a database allows insertion of an order with a customer ID that does not exist in the Customer table. Which database concept is violated? Also explain two possible problems this may create.
- Consider attribute Address="House#12, Street 5, Peshawar". Discuss whether Address should be treated as atomic attribute or composite attribute.
- A student says "Primary keys and candidate keys are exactly the same." Critically analyze this statement with suitable examples.

[15 Minutes, 3 + 4 + 3 marks]

2. A hospital database stores:

Patient(PatientID, Name, DischargeDate)

Currently admitted patients have NULL in 'DischargeDate'. Explain what this NULL value represents and why it is conceptually acceptable.

[10 Minutes, 5 marks]

CLO #2: Design a conceptual model using (E)ER diagrams for an enterprise

3. A hospital wants to maintain a database system for managing its operations.

- The hospital has multiple departments. Each department is identified by a unique Department ID and stores its name, floor number, and contact extension. Every department is managed by exactly one doctor, while a doctor may manage at most one department.
- Doctors work in one department only. For each doctor, the hospital stores doctor ID, name, specialization, qualification, salary, and multiple phone numbers.
- Patients are registered with a unique patient ID. The system stores patient name, gender, age, blood group, address, and emergency contact details.
- A patient may visit the hospital multiple times. During each visit, the patient consults exactly one doctor. For every visit, the system stores visit date, symptoms, diagnosis, and prescribed medicines.
- Some patients are admitted to hospital wards. Each ward contains multiple rooms, and each room contains multiple beds. Bed numbers are unique only within a room. The system keeps track of admission date and discharge date.
- Laboratory tests can be prescribed during visits. Each test has a test ID, test name, cost, and result status. A visit may generate multiple tests.

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Design a complete ERD including:

- Entities
- Attributes
- Primary keys
- Relationships
- Cardinalities
- Participation constraints
- Weak entities (if any)

[20 Minutes, 15 marks]

4. A manufacturing company maintains information about suppliers, parts, and projects.

- Suppliers provide parts for projects.
 - A supplier may supply many parts.
 - A project may require many parts.
 - The same part may be supplied by different suppliers for different projects.
 - For each supply transaction, the quantity supplied is recorded.
- a) Design an ER diagram for the scenario using Multiple Binary Relationship
b) Now model the scenario using Ternary Relationship
c) Explain what information may be lost if only binary relationships are used.

[15 Minutes, 10 marks]

CLO #3: Develop a normalized relational design to remove anomalies in a set of relations

5. Consider the following relation used in a software company database:

ProjectAssignment(EmpID, EmpName, DeptID, DeptName, ProjectID, ProjectName, ClientID, ClientName, ManagerID, ManagerName, HoursWorked, HourlyRate)

- a) Identify all functional dependencies present in the relation.
b) Determine the highest normal form satisfied by the relation. Justify your answer.
c) Normalize the relation up to Third Normal Form (3NF). Clearly show the resulting relations along with their Primary Keys and Foreign Keys.

[15 Minutes, 15 marks]

CLO #4: Implement the database schema developed against the designed conceptual model.

6. A student creates the following sparse index:

Key	Pointer
100	P1
130	P2
170	P3

However, the actual file blocks are unordered:

- B1 → 130, 105, 100
- B2 → 170, 120, 140
- B3 → 200, 180, 160

Analyze this structure critically.

[10 Minutes, 5 marks]

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7. Insert the following keys into a B-tree

15, 25, 5, 10, 20, 30, 17, 45, 29

$m=4$

Show final structure of the tree. Also explain why B+ tree may require more node splits for the same data.
[15 Minutes, 15 marks]

8. Bitmap indices are widely used in data warehousing systems but not always suitable for transactional systems.

- Explain why bitmap indices are efficient for low-cardinality attributes.
- Analyze why they become inefficient for high-cardinality attributes.
- Discuss their impact on:
 - storage space
 - update operations
- Suggest a scenario where bitmap indices are clearly preferable over B+ tree indexing.

[10 Minutes, 2 + 2 + 3 + 3 marks]

CLO #5: Author and use queries using relational algebra and SQL.

9. Consider the following bank database, where the primary keys are underlined.

branch(branch-name, branch_city, assets)

customer(customer-CNIC, customer_name, customer_street, customer_city)

loan(loan-number, branch_name, amount)

borrower(customer-CNIC, loan-number)

account(account-number, branch_name, account_holder_CNIC, balance)

Construct the following SQL queries for this relational database.

- Find all customers who have an account at any of the branches located in "Islamabad".
- Delete borrower records for customers who have loans but no account anywhere in the bank.
- Find the names of customers who have accounts in more than one city.
- Find branches where the average account balance is higher than the average loan amount.
- Decrease the assets of branches by the total loan amount issued from that branch.

[25 Minutes, 20 marks]

10. Consider a university course registration system having the following relations:

Student(s_id, name, dept)

Course(c_id, title, credits)

Enroll(s_id, c_id)

The database statistics shows the following:

- Student: 20,000 records
- Course: 1,000 records
- Enroll: 100,000 records
- Only 200 courses have credits = 3
- Each course has ~100 students enrolled

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Suppose the following query is being run on the database:

```
SELECT Student.name  
FROM Student, Course, Enroll  
WHERE Student.s_id = Enroll.s_id  
      AND Course.c_id = Enroll.c_id  
      AND Course.credits = 3;
```

- a) Construct the initial query tree (Cartesian products + selections).
- b) Transform it into an optimized query tree by:
 - i. pushing selections down
 - ii. reordering joins
- c) Estimate intermediate result sizes for:
 - i. initial plan
 - ii. optimized plan
- d) Suggest an efficient evaluation strategy (execution plan) using indexes if available.

Explain how join ordering impacts performance.

[20 Minutes, 15 marks]